

Route Correction and Variance Decomposition for Aerobic Decoupling: A 13K-Workout Study of Durability Dimensionality

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Millions of athletes use the speed-based Decoupling Index (DI) displayed by GPS-watch platforms such as TrainingPeaks and Strava, yet its sensitivity to route geometry has never been quantified. This study analysed $N = 13,395$ real-world workouts from $K = 674$ users to examine whether route terrain—rather than physiological cardiac drift—dominates speed-based DI, and to test the empirical dimensionality of three of the four constructs in the durability framework proposed by Meixner et al. (2025). Three speed-based indices were defined: DI (durability), the Fading Index (FI; fatigability), and the Resilience Index (RI; resilience); repeatability was not operationalized. A controlled simulation ($N = 5,000$ synthetic workouts with zero cardiac drift) showed that route geometry alone produces DI values ranging from 0.70 to 1.40 (cross-validated $R^2 = 0.80$). In real data, route features predicted DI with $R^2 = 0.49$; after Gradient-Adjusted Cardiac Drift (GACD) regression, this predictability was eliminated ($R^2 \approx 0.02$). Notably, 42% of these hilly routes were front-climb profiles that mechanically produce $DI < 1.0$ even with zero cardiac drift. PCA on person-level medians retained a single factor: the second eigenvalue (1.004) fell below the parallel-analysis threshold (1.036), and bootstrap resampling over users ($B = 2,000$) retained a second factor in only 5.3% of replicates (margin 95% CI $[-0.103, +0.011]$). DI and FI were redundant ($r = -0.60$), and RI was too unreliable (ICC = 0.10) to be interpreted as a distinct construct; a PCA restricted to DI and FI likewise retained one factor explaining 80.0% of variance. Under speed-based operationalization the three measured constructs therefore do not separate into distinct empirical dimensions. Variance decomposition revealed that day-to-day Occasion variance dominates all metrics (56–84%), necessitating multi-session aggregation: gradient sensitivity from the GACD regression reached adequate reliability (Spearman–Brown ≥ 0.80) at $k \geq 5$ sessions, a guideline validated primarily in the cycling-dominated high-engagement subset. Sport-specific analysis confirmed that the Occasion-dominant pattern holds for cycling (Occasion 50–78% across the three GACD-derived metrics); the running subsample was too small for stable estimation. These findings demonstrate that speed-based DI is heavily contaminated by route geometry and that gradient-adjusted, multi-session profiling provides a more *reproducible* individual signal—reliability, not criterion validity—for longitudinal monitoring.

aerobic decoupling | durability | cardiac drift | heart rate | gradient sensitivity
| measurement artifact | field testing

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Introduction

Aerobic decoupling—the progressive rise in heart rate (HR) relative to pace or power during prolonged exercise—has been widely adopted as a field-based indicator of endurance fitness (4, 5). Platforms such as TrainingPeaks and Strava present the Decoupling Index (DI) to millions of athletes, applying Friel’s $< 5\%$ threshold to indicate adequate aerobic base. This threshold has, to our knowledge, never been empirically validated in a peer-reviewed study. The most closely related large-scale study is Smyth et al. (18), who characterised internal-to-external workload decoupling in 82,303 marathon runners and demonstrated that both decoupling magnitude and onset are associated with performance. Their work establishes the ecological validity of field-based decoupling measurement at scale, but did not examine the route-geometry artifact, the dimensional structure of multiple decoupling indices, or the variance attributable to person, route, and occasion—the gaps addressed here. The closest methodological precursor is the cardiac cost index (HR divided by running velocity) advocated by Billat et al. (8) as a drift measure that is sensitive to pacing; like DI, however, it normalises HR by speed alone and therefore inherits the same exposure to terrain. Bonomi, Papini, and Sartor (9) modelled instantaneous HR drift during laboratory running so that corrected HR better tracked VO_2 kinetics (77% reduction in submaximal hysteresis AUC); that correction treats drift as contamination of the HR signal itself, whereas GACD residualises field HR against gradient and speed in order to isolate the time-dependent drift term and does not address route-geometry contamination of speed-based DI. That terrain is not incidental to durability was shown by Lloria-Varela et al. (11), who found in trail runners that a 90-minute submaximal effort produced progressive slowing of uphill segments at stable HR—internal-to-external decoupling driven by gradient rather than by time alone. Their protocol was controlled and terrain-matched by design; the present study asks what happens to the same signal when terrain is uncontrolled, as it is for the overwhelming majority of GPS-tracked training.

On variable terrain, speed—determined primarily by gradient—is an imperfect proxy for external workload, introducing a route-geometry component into speed-based DI. What has not been established is (a) the magnitude of this component, (b) whether it dominates physiological signal in real-world data, and (c) what meaningful signal remains once

it is corrected.

Meixner et al. (13) proposed four conceptual constructs for endurance durability: durability, fatigability, repeatability, and resilience. The accompanying commentaries—Colosio et al. (3) on physiological distinctions, Faude et al. (25) on methodological gaps, Lucernoni et al. (26) on reconceptualising resilience, and the authors' rebuttal (27)—reveal an active debate over whether these four constructs are empirically separable, with no consensus yet reached. Separately, Jones (24) proposed physiological resilience as a fourth independent determinant of endurance performance, providing theoretical grounding for the resilience construct. All discussions have remained at the conceptual level. To bridge theory and large-scale field data, this study introduces three author-defined operationalizations of Meixner et al.'s constructs: the Decoupling Index (DI; operationalizing durability), the Fad-ing Index (FI; fatigability), and the Resilience Index (RI; resilience). The fourth construct (repeatability) was not operationalized in this study due to the absence of repeated-bout protocols in the FitRec dataset. These indices are not defined in Meixner et al. (13); they are original operationalizations designed to map the theoretical constructs to measurable GPS and HR field data. One likely reason no prior large-scale study exists is a disciplinary gap: Meixner et al.'s framework defines DI as HR/Power, while the FitRec dataset (15)—published in the recommender-systems community—contains HR and GPS but no power data. Yet the majority of real-world users compute HR/Speed via consumer GPS watches. This study does not challenge the theoretical constructs themselves; rather, it examines the empirical structure that emerges when those constructs are operationalized with the speed-based metrics that millions of users actually employ, a necessity for field-based durability research (10). While Friel (2009) (5) proposed a 5% DI threshold for physiological decoupling, the present results suggest that on variable terrain, this index is heavily contaminated by route-geometry artifacts, necessitating a more robust gradient-adjusted approach. This distinction—between theoretical construct validity and field-level operationalization—is the central concern of the present work.

The reliability of field-based HR metrics has long been a concern (7, 20). The IOC consensus (2) emphasized the external-load/internal-response distinction, but no study has quantified the variance attributable to person, route, and occasion in HR-response metrics.

Research Questions. All analyses use speed-based DI (HR/Speed), not power-based DI (HR/Power), because no power data are available in FitRec. Results apply to the majority of real-world users who compute DI via consumer GPS watches. It should be noted that HR/Speed is a pragmatic but theoretically limited proxy for cardiac efficiency in cycling—where speed is determined by power output, gradient, aerodynamic drag, and rolling resistance—and a stronger proxy in running, where speed is more directly related to metabolic cost on flat terrain. The implications of this distinction for the cycling-dominated dataset (62.0%) are addressed in the Discussion.

CRQ1: What is the empirically supported dimensionality of the durability framework under speed-based DI?

- H1** DI and FI collapse onto a single construct, indicating that the speed-based operationalizations do not separate durability from fatigability.
- H2** All durability dimension ICCs fall below 0.50, indicating insufficient reliability for four independent dimensions.
- H3** DI route predictability vanishes after gradient correction (GACD), confirming that the route-geometry component can be effectively removed.

CRQ2: To what extent do stable individual physiology, route properties, and transient daily state each account for the variance in HR-response metrics?

- H4** Transient daily state (Occasion) accounts for > 50% of variance, necessitating multi-session aggregation for reliable measurement.

Three exploratory hypotheses were also examined:

- E1** Temporal placement of descent affects cardiac drift.
- E2** A more stable HR parameter than DI exists.
- E3** ICC $\ll$ split-half reliability (rank-order stability > absolute consistency).

Contributions. **C1** Evidence that the three operationalizable constructs (DI, FI, RI; repeatability was not operationalized) do not separate empirically under speed-based field metrics (PCA retained one factor; PC2 = 1.004 vs. threshold 1.036, retained in only 5.3% of bootstrap replicates), with quantification of the route-geometry artifact ($R^2 = 0.80$; 2-fold range) and demonstration that naive power estimation overcorrects. **C2** A large-scale variance decomposition (Person/Route/Occasion) for HR-response metrics: the same person on the same route shows 26–54% day-to-day variation, indicating that these speed-based operationalizations require multi-session aggregation to be reliably measured. **C3** A between-person association between descent placement and cardiac drift, with modest within-person support. **C4** Convergence analysis showing that gradient sensitivity requires ≥ 5 sessions for adequate reliability.

Methods

Data and Filtering. The FitRec dataset (15) was used, comprising 253,020 GPS-tracked workouts from Endomondo. Each workout record j contains irregularly time-stamped arrays of heart rate ($HR_{j,i}$, optical wrist sensor, bpm), speed ($v_{j,i}$, GPS-derived, $\text{km}\cdot\text{h}^{-1}$ in the raw data), altitude ($a_{j,i}$, barometric or GPS-derived, m), and geographic coordinates, where $i = 1, \dots, N_j$ indexes the data points. Where the recorded speed array was missing or shorter than 10 points, speed was derived from GPS coordinates via the Haversine formula (values exceeding $150\text{ km}\cdot\text{h}^{-1}$ were excluded as

outliers). Of the 13,395 included workouts, 11,098 (82.9%) used Haversine-derived speed and 2,297 (17.1%) retained a recorded GPS speed array; the GACD analyses in Studies 2–3 are restricted to the recorded-speed subset. The high Haversine share reflects incomplete speed arrays in the public Endomondo/FitRec dump (the speed field was often short or missing), not a property of contemporary GPS watches. The recorded-speed subset is more cycling-heavy (83.9% vs. 62.0% in the full sample); the Study 2 simulation is nonetheless parameterised from the full-sample distributions and therefore speaks to the broader hilly inclusion set. The gender variable indicates 93.9% male, 4.9% female, 1.2% unknown; age, body mass, and training history are unavailable. Sensor make and firmware version are not recorded.

A four-step inclusion pipeline was applied: (1) HR, altitude, and speed arrays each required $N_j \geq 30$ finite data points (a minimum for stable within-half mean estimation with ≥ 15 points per half); missing values were linearly interpolated between flanking valid points; (2) total duration $T_j = t_{N_j} - t_1 > 90$ min, ensuring sufficient exercise duration for cardiac drift to manifest (4); (3) altitude range $\Delta a_j = \max_i(a_i) - \min_i(a_i) > 200$ m, a conservative threshold for ensuring sufficient gradient variation to confound the speed-based DI (approximately the lower quartile of the retained dataset; median = 349 m). To address sensor-induced barometric drift, a workout was treated as a geographic loop if the Haversine distance between its first and last GPS coordinates was < 100 m (requiring ≥ 100 coordinate points); such loops were excluded when the start-to-end altitude deviation satisfied $|\Delta a_{\text{drift}}| = |a_{N_j} - a_1| > 50$ m, which is inconsistent with a closed route and indicates barometric (or GPS-altitude) drift. This filter is a pragmatic screen rather than a calibrated model of sensor drift rates. (4) each temporal half (H1: $i \leq \lfloor N_j/2 \rfloor$; H2: $i > \lfloor N_j/2 \rfloor$) required > 5 points with $v_i > 0.5 \text{ km}\cdot\text{h}^{-1}$ (i.e., ≥ 6 per half), where $0.5 \text{ km}\cdot\text{h}^{-1}$ excludes stationary GPS noise while preserving low-speed segments on steep climbs. When speed was entirely unavailable, a heart-rate-only DI ($= \bar{\text{HR}}_{\text{H2}}/\bar{\text{HR}}_{\text{H1}}$) was computed as a fallback; FI and RI required speed.

This yielded $N = 13,395$ workouts from $K = 674$ users for Study 1 (cycling 62.0%, running 16.9%, mountain biking 16.6%, other 4.5%; median duration 180 min). An earlier snapshot of the filtered file contained 13,750 workouts from 675 users; regenerating from the public FitRec JSON with the current scripts—including the loop barometric-drift filter described above—reproducibly yields 13,395 from 674 users, which is the sample used throughout. Studies 2–3 required ≥ 5 repeated workouts per user *and* a recorded (not Haversine-imputed) speed array of sufficient length for the GACD regression, yielding a high-engagement subset of $N = 2,297$ workouts from $K = 309$ users (cycling 83.9%, running 5.4%, mountain biking 9.7%, other 1.0%; median duration 179 min [IQR 131–261]).

Metric Definitions. Speed in the raw data is recorded in $\text{km}\cdot\text{h}^{-1}$ (metrics 1–3). For the within-workout regression (metrics 4–6), speed was converted to $\text{m}\cdot\text{s}^{-1}$ ($\div 3.6$); the gradient computation uses speed in $\text{m}\cdot\text{s}^{-1}$ and timestamps in

seconds.

(1) Decoupling Index (DI). Let $\mathcal{S}_h = \{i \in \text{H}_h : v_i > 0.5 \text{ km}\cdot\text{h}^{-1}\}$ for $h \in \{1, 2\}$. The DI is:

$$\text{DI} = \frac{\bar{\text{HR}}_{\mathcal{S}_2} / \bar{v}_{\mathcal{S}_2}}{\bar{\text{HR}}_{\mathcal{S}_1} / \bar{v}_{\mathcal{S}_1}}$$

where $\bar{\text{HR}}_{\mathcal{S}_h} = |\mathcal{S}_h|^{-1} \sum_{i \in \mathcal{S}_h} \text{HR}_i$ and analogously for $\bar{v}$. DI > 1 indicates cardiac drift.

(2) Fading Index (FI). For FI, point-to-point gradient was computed from the cumulative horizontal distance: $D_i = \sum_{l=1}^i v_l \cdot \Delta t_l / 3600 \times 1000$ (m), with minimum horizontal step 0.1 m, and $g_i = \text{clip}(100 \times \Delta a_i / \Delta D_i, -50, +50)\%$. The gradient range was partitioned into five bins $\mathcal{B}_b$: $[-50, -10)$, $[-10, -3)$, $[-3, 3)$, $[3, 10)$, $[10, 50)\%$, following conventional gradient classifications (flat: $< 3\%$; moderate: $3\text{--}10\%$; steep: $> 10\%$). The boundary thresholds reflect standard terrain boundaries used by consumer training platforms; sensitivity to these bin definitions was not formally tested, though the overall DI–FI correlation proved robust across pre-filtering thresholds (see below).

$$\text{FI}_b = \frac{\bar{v}_{\mathcal{S}_2 \cap \mathcal{B}_b}}{\bar{v}_{\mathcal{S}_1 \cap \mathcal{B}_b}}, \quad \text{FI} = \frac{1}{|\mathcal{B}^*|} \sum_{b \in \mathcal{B}^*} \text{FI}_b$$

where $\mathcal{B}^* = \{b : |\mathcal{S}_1 \cap \mathcal{B}_b| > 3 \text{ and } |\mathcal{S}_2 \cap \mathcal{B}_b| > 3\}$. The rationale for operationalizing fatigability as speed decline across gradient-matched bins rather than unbinned speed decay over time is construct isolation: comparing second-half to first-half speed on flat terrain, or on climbs of similar grade, controls for route geometry and captures the athlete’s inability to maintain output for a given terrain demand. A decline in FI (< 1.0) thus reflects speed decay that cannot be attributed to shifting gradient, which provides a field operationalization of neuromuscular or metabolic fatigability. On average, 4.2 ± 0.8 of the five bins were populated per workout (> 3 points per half), and 82.4% of workouts had at least four bins populated. Of the $N = 13,395$ included workouts, 13,387 (99.9%) had at least one gradient bin populated in both halves; FI was undefined for the remaining 8 workouts.

(3) Resilience Index (RI). The RI operationalizes the resilience construct proposed by Jones (24), who identified physiological resilience as a fourth independent determinant of endurance performance. This operationalization is a preliminary proxy: Jones (2024) defines resilience in terms of preservation of physiological profiling variables (VO_2max , economy, critical power) under fatigue, whereas RI captures only speed maintenance between the first and last climb—a gap that future studies with laboratory-grade measurements could address. Three additional confounds are noted: (a) the relative steepness of C_1 vs. C_m is not controlled; (b) temporal position of climbs within the workout is not normalised; (c) RI values are not directly comparable across workouts with different numbers of climbs m . Altitude was smoothed by a trailing moving average (window $w = \max(1, \lfloor N/20 \rfloor)$, mode = “valid”), yielding a smoothing window of approximately 5% of the workout duration; this balances noise reduction against oversmoothing of true elevation changes. A

climb was initiated when the smoothed altitude increased by > 0.5 m from one point to the next, and terminated when this condition ceased; only climbs with cumulative vertical gain > 50 m were retained. This threshold is an author-defined pragmatic choice following a standard cycling industry convention of > 50 m vertical gain to delimit a climb; it is not a peer-reviewed standard, and sensitivity to this threshold was not formally assessed. Let $C_1, \dots, C_m$ denote the retained climbs. If $m \geq 2$:

$$RI = \bar{v}_{C_m} / \bar{v}_{C_1}$$

where $\bar{v}_{C_l} = |\{i \in C_l : v_i > 0.5\}|^{-1} \sum_{i \in C_l, v_i > 0.5} v_i$. Workouts with $m < 2$ received no RI.

(4–6) Within-workout regression. For the regression-derived metrics, speed was in $\text{m} \cdot \text{s}^{-1}$ and gradient was computed as:

$$g_i = \text{clip}\left(100 \times \frac{a_{i+1} - a_i}{d_i}, -50, +50\right),$$

$$d_i = \max(v_i \cdot \Delta t_i, 0.1 \text{ m}), \quad \Delta t_i = \max(t_{i+1} - t_i, 0.1 \text{ s})$$

Note that the minimum distance floor is 0.1 m (not 0.5 m as in FI), reflecting the finer resolution needed for regression. All variables were evaluated at midpoints: $HR_i^* = (HR_i + HR_{i+1})/2$, $v_i^* = (v_i + v_{i+1})/2$, $t_i^* = (t_i + t_{i+1})/2$ (min). On the subset of valid indices ($|\mathcal{V}| \geq 20$ required), the OLS regression:

$$HR_i^* = \beta_0 + \beta_{\text{grad}} g_i + \beta_{\text{speed}} v_i^* + \beta_{\text{time}} t_i^* + \varepsilon_i, \quad (1)$$

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Linearity was confirmed via residual analysis, with homoscedasticity observed across the primary speed and gradient ranges. Importantly, β_{time} captures the within-session rate of HR drift after controlling for gradient and speed; this should be distinguished from the between-session durability construct defined by Maunder et al. (12), which concerns shifts in the power–duration relationship after a fatiguing bout.

Sensitivity Analysis: Temporal Autocorrelation. Point-level GPS data exhibit temporal autocorrelation, which violates the OLS assumption of independent errors. To quantify its impact, a sensitivity analysis was conducted on a random subsample of $N = 300$ workouts (median $n = 499$ data points per workout). For each workout, the GACD regression was re-run and the following diagnostics computed:

Durbin–Watson (DW) statistics. The DW statistic was computed as $DW = \sum_{t=2}^n (e_t - e_{t-1})^2 / \sum_{t=1}^n e_t^2$, where e_t are OLS residuals. A value of 2.0 indicates no autocorrelation. The median DW was 0.376 (IQR: 0.254–0.513), and 100% of workouts showed strong positive autocorrelation ($DW < 1.5$; Supplementary Figure S2a).

Autocorrelation function (ACF). The lag-1 autocorrelation of GACD residuals was high (median $\rho_1 = 0.797$, IQR: 0.722–0.860; Supplementary Figure S2b), with substantial autocorrelation persisting to lag 10+. Under an AR(1) approximation, the effective sample size is reduced by a median of 89% ($n_{\text{eff}}/n = 0.113$).

Newey–West HAC standard errors. To assess the impact on inference, Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors were computed with a Bartlett kernel and bandwidth $L = \lfloor 4(n/100)^{2/9} \rfloor$. The HAC standard errors were approximately $2\times$ larger than OLS standard errors across all coefficients (median SE ratio: β_{gradient} : 1.98; β_{time} : 2.10; β_{speed} : 2.01). HAC-corrected interval estimates are reported rather than the ratio alone. For the median workout, the 95% CI half-width widened from 0.361 to 0.753 bpm/% for β_{gradient} , from 0.021 to 0.043 bpm/min for β_{time} , and from 0.620 to 1.278 for β_{speed} . The corresponding HAC intervals around the median coefficient are +1.80 [+1.05, +2.56] bpm/% for gradient, +0.021 [−0.022, +0.064] bpm/min for time, and −0.009 [−1.29, +1.27] for speed. The share of workouts whose coefficient remained individually significant at $\alpha = .05$ fell from 95.0% to 85.0% for β_{gradient} , from 79.0% to 59.7% for β_{time} , and from 74.3% to 51.3% for β_{speed} . Because the $\sim 2\times$ SE inflation is itself a lower bound under this persistence regime (28), these significance-retention rates are *upper bounds*: the true rates under a more conservative long-run variance estimator could be lower. Gradient sensitivity is nevertheless the only coefficient that remains individually significant in the large majority of sessions even under this caveat, whereas a single session’s drift rate or speed coefficient is frequently indistinguishable from zero once autocorrelation is accounted for.

Cochrane–Orcutt GLS (AR(1)) was applied to the same 300-workout subsample to check whether OLS point estimates are interchangeable with a more efficient estimator. Rank-order agreement was high (β_{gradient} : $r = 0.83$; β_{time} : $r = 0.86$; β_{speed} : $r = 0.76$), but GLS magnitudes were substantially attenuated (median β_{gradient} +1.70 vs. +0.24 bpm/%, a ~ 7 -fold reduction). That attenuation is expected and quantitatively consistent with the observed persistence. Under AR(1) GLS with ρ in both the error and a similarly persistent regressor, the quasi-differenced series $(g_t - \rho g_{t-1})$ has variance approximately $(1 - \rho)^2 / (1 - \rho^2)$ that of the raw gradient; with $\rho \approx 0.80$ this factor is ≈ 0.11 (~ 9 -fold). That formula assumes the same ρ for the regressor and the error; if gradient lag-1 autocorrelation differs from residual $\rho_1 = 0.797$, the expected factor changes, which can account for the remaining gap between ~ 9 -fold and the observed ~ 7 -fold shrinkage ($1.70/0.24 \approx 7.1$). The GLS coefficient therefore estimates the HR response to short-run gradient *changes* after the persistent terrain level has been differenced out, not the contemporaneous terrain–HR slope that OLS targets. GLS is not a substitute estimator for the profiling coefficients. OLS is retained because the quantity of interest is that contemporaneous slope. Rank-order agreement of $r = 0.83$ means about 31% of between-person variance is not shared: OLS and GLS would disagree on the ordering of a non-negligible minority of athlete pairs (on the order of $1 - r \approx 17\%$). Practitioners using gradient sensitivity for individual comparisons should treat OLS ranks as the intended contemporaneous-slope ranking, not as interchangeable with a GLS ranking.

The primary quantity of interest for individual profiling is

the OLS point estimate of the regression coefficients rather than inference on individual session slopes. Per-workout R^2 values should be interpreted as upper-bound estimates of model fit. where $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times 4}$ (intercept, g , v^* , t^*) yielded: (4) GACD ($\hat{\beta}_{\text{time}}$, bpm·min $^{-1}$); (5) Gradient sensitivity ($\hat{\beta}_{\text{grad}}$, bpm·% $^{-1}$); (6) Speed sensitivity ($\hat{\beta}_{\text{speed}}$, bpm·(m·s $^{-1}$) $^{-1}$). Per-workout $R^2 = 1 - \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 / \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$. Point-level gradients were clipped to $\pm 50\%$, as values beyond this range exceed the steepest rideable/runnable terrain and indicate sensor artifacts. Although gradient (g_i) and speed (v_i^*) exhibit collinearity in field environments, multicollinearity was minimal (median VIF: gradient 2.01, speed 2.09, time 1.07), with 99.8% of sessions falling below the VIF < 10 threshold (92.4% below 5) (6).

Estimated Power Experiment. DI was recomputed with the denominator replaced by estimated metabolic power $P_i = v_i \cdot C_w(g_i)$ (W·kg $^{-1}$), where $C_w(g)$ is the Minetti (14) gradient cost of walking transport (J·kg $^{-1}$ ·m $^{-1}$), computed from the published 5th-order polynomial (Eq. 2 of Minetti et al., 2002):

$$C_w(i) = 280.5i^5 - 58.7i^4 - 76.8i^3 + 51.9i^2 + 19.6i + 2.5$$

where i is gradient as a fraction (e.g., $i = 0.10$ for 10%), valid for $i \in [-0.45, +0.45]$. Values were floored at 0.1 J·kg $^{-1}$ ·m $^{-1}$ to prevent near-zero denominators on moderate descents:

$$\text{DI}_{\text{EstP}} = \frac{\overline{\text{HR}}_{S_2} / \bar{P}_{S_2}}{\overline{\text{HR}}_{S_1} / \bar{P}_{S_1}}$$

The metabolic cost of transport was estimated using the 5th-order polynomial published by Minetti et al. (2002) (14). To ensure robust estimation, extreme outliers were removed using 1st and 99th percentile trimming (23), resulting in a final sample of $N = 13,080$ for the power-estimation experiments. Prolonged exercise was defined as $T > 90$ min, consistent with the manifesting period for cardiovascular drift (4, 19). Note that the Minetti cost function was originally derived for human walking and running; its application to a dataset dominated by cycling (62.0%) introduces an inherent bioenergetic mismatch (see Discussion).

Study 1: Construct Validity — H1 (DI–FI redundancy), H2 (ICC < 0.50). To test whether the proposed four dimensions are empirically distinct (H1) and individually reliable (H2), person-level medians of DI, FI, and RI were computed for each user with ≥ 5 workouts ($N = 13,395$; for RI, $N = 9,680$). Median-based aggregation was selected due to high Kurtosis (e.g., 590 for FI) indicating extreme outliers and heavy-tailed distributions. Pairwise Pearson correlations were estimated via nonparametric bootstrap ($B = 2,000$; users resampled with replacement; 95% CI from 2.5th and 97.5th percentiles). If H1 holds, DI and FI should show a strong negative correlation (FI < 1 when DI > 1) and PCA should retain only one factor.

PCA was performed on column-standardised person-level medians ($z_{ij} = (x_{ij} - \bar{x}_j) / s_j$) for users with valid values on

all three metrics. Factor retention was determined by parallel analysis (21): observed eigenvalues were compared to the 95th percentile of eigenvalues from $B = 1,000$ column-wise random permutations of standard normal data of the same dimensions. Because a retention decision based on a point estimate is uninformative when an eigenvalue lies close to its threshold, the sampling uncertainty of each eigenvalue was quantified by nonparametric bootstrap over users ($B = 2,000$; users resampled with replacement). For each replicate the eigenvalues and the eigenvalue-minus-threshold margins were recomputed, yielding percentile 95% CIs and the proportion of replicates in which each component was retained. Because RI proved unreliable (ICC = 0.10; see Study 3), the PCA was repeated on DI and FI alone as a robustness check, since loadings of an unreliable variable are dominated by measurement error. ICC(1,1) was selected as the primary reliability metric because workouts are unique, non-overlapping samples per user and cannot be treated as crossed or fixed factors; thus, 2-way models (ICC(2,1), ICC(3,1)) are inappropriate for this dataset. ICC(1,1) was computed for each metric on users with ≥ 5 workouts (see Study 3); H2 predicts all ICCs < 0.50. For context, in field-based sports science monitoring, ICC > 0.70 is typically required for acceptable individual tracking (2, 20), and laboratory-based durability measures frequently achieve ICC 0.70–0.85 (e.g., peak speed change in Malinen et al., 2025; HR drift in controlled protocols). The lower values observed here (0.20–0.45) reflect the unconstrained field environment.

The three-way variance decomposition partitions total observed variance into Person (σ_P^2), Route (σ_R^2), and Occasion (σ_O^2) components. The model assumes additivity: total variance $\sigma_T^2 = \sigma_P^2 + \sigma_R^2 + \sigma_O^2$, where the Person and Route factors are orthogonal. Under this assumption, the Person share is given directly by the unadjusted ICC(1,1) (%Person = ICC $\times$ 100). The Route share is estimated from the route-prediction cross-validated R^2 on within-person residuals (%Route = $R_{\text{route}}^2 \times (1 - \text{ICC}) \times 100$), and the remaining variance is assigned to Occasion (%Occasion = 100 – %Person – %Route). The route-matched ICC analysis (Study 3) provides a direct empirical test of this additivity assumption: by calculating ICC on workout pairs from the same user on the same route ($N = 450$ pairs, 93 users), route variance is held constant, and the increase in ICC over the unadjusted baseline measures the route component directly.

Robustness check. To ensure that the findings were not artifacts of specific inclusion thresholds, a sensitivity analysis was performed by varying $T_{\min}$ from 60 to 120 min and Δa from 100 to 300 m. Across all 25 combinations, the DI–FI correlation remained remarkably stable ($\bar{r} = -0.60$, SD = 0.02, range $[-0.63, -0.57]$), confirming that the observed redundancy is a robust phenomenon independent of specific filtering choices.

Study 2: Route-Geometry Correction — H3 (gradient correction), E1 (descent placement). Route prediction. Let $\mathbf{x}_j \in \mathbb{R}^{13}$ denote 13 route features (total ascent/descent, altitude range, max/min altitude, gradient mean/std, climb/descent/flat fractions, ascent-front and

descent-front ratios, duration). Within each cross-validation fold f , features were z -standardised using training-fold statistics only ($\mu_p^{(f)}$, $\sigma_p^{(f)}$ computed on train_f ; applied to both train_f and test_f), and RidgeCV regression (selecting $\alpha \in \{0.01, 0.1, 1.0, 10, 100\}$ via internal leave-one-out CV) was fitted:

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}, \quad \alpha^* = \arg \min_{\alpha} \text{LOO-CV}(\alpha)$$

and gradient boosted trees (100 trees, max depth 3) were fitted. Generalisation was assessed by group k -fold cross-validated R^2 ($k = \min(5, K_{\text{valid}})$; all workouts of each user in the same fold):

$$R_{\text{CV}}^2 = 1 - \frac{\sum_{f=1}^k \sum_{j \in \text{test}_f} (y_j - \hat{y}_j^{(f)})^2}{\sum_{f=1}^k \sum_{j \in \text{test}_f} (y_j - \bar{y})^2}$$

The hilly subset was restricted to $\Delta a_j > 400$ m. For GACD, the target was the within-person deviation $\tilde{y}_{kj} = y_{kj} - \bar{y}_k$ (users with ≥ 3 workouts), isolating route effects from person-level means.

Empirically-parameterised controlled experiment. $N = 5,000$ synthetic workouts ($N_{\text{pts}} = 60$, split at 30) were generated with cardiac drift set exactly to zero. All distributional parameters were derived from the FitRec dataset. Three route types were sampled with empirical proportions (classified by ascent-front ratio from `abc_metrics.csv`, $N = 2,297$):

- Front-climb ($\text{asc_front} > 0.6$; 42.2%): $g_{\text{H1}} \sim \mathcal{N}(+1.0, 1.84^2)$, $g_{\text{H2}} \sim \mathcal{N}(-1.0, 1.84^2)$
- Back-climb ($\text{asc_front} < 0.4$; 10.2%): reversed
- Symmetric ($0.4 \leq \text{asc_front} \leq 0.6$; 47.6%): $g \sim \mathcal{N}(0, 1.84^2)$ throughout

The gradient asymmetry ($\pm 1.0\%$) was derived from the ascent-front distribution (front-climb mean ≈ 0.69), and the point-level gradient standard deviation (1.84%) was chosen to represent moderate terrain. The actual per-workout gradient standard deviation in the dataset is larger (mean = 4.96%). The simulation is therefore a *conservative lower bound* on route-geometry bias: using the smaller SD understates, rather than overstates, the DI range that geometry alone can produce. A sensitivity run with the empirical SD would be expected to widen the simulated DI range beyond 0.70–1.40. Per-workout physiological parameters were sampled from FitRec-derived distributions: $\text{HR}_0 \sim U(107, 159.5)$ (P5–P95 of `avg_hr` in the full dataset, $N = 13,395$), $\gamma_{\text{HR}} \sim \max(0.1, \mathcal{N}(1.91, 1.48^2))$ bpm/% (from `gacd_gradient_coef`), $v_0 \sim \max(1.0, \mathcal{N}(1.91, 0.44^2))$ m/s, $\gamma_v \sim \max(0.01, \mathcal{N}(0.27, 0.15^2))$ (m/s)/% (derived from Minetti (14) cost function gradients). The data-generating process was:

$$\begin{aligned} \text{HR}_i &= \text{HR}_0 + \gamma_{\text{HR}} \cdot g_i + \epsilon_i^{\text{HR}}, \quad \epsilon_i^{\text{HR}} \sim \mathcal{N}(0, 3^2) \\ v_i &= \max(0.5, v_0 - \gamma_v \cdot g_i + \epsilon_i^v), \quad \epsilon_i^v \sim \mathcal{N}(0, 0.3^2) \end{aligned}$$

No time-dependent term was included; any non-unity DI arises purely from gradient asymmetry between halves. The only experimentally controlled variable is cardiac drift = 0; all other parameters reflect empirical distributions from the dataset under study.

Descent-placement analysis. Define the descent-front ratio:

$$\delta = \frac{\sum_{i \in \text{H1}} |\min(\Delta a_i, 0)|}{\sum_{i=1}^N |\min(\Delta a_i, 0)|}$$

This was tested as a predictor of GACD via: (a) between-person Pearson r (and separately controlling for total descent); (b) within-person Pearson r for users with ≥ 5 varied routes ($n = 108$); (c) steepness stratification ($\sigma_g < 5$; 5–8; > 8); (d) sport-specific subgroups. All eight exploratory p -values were adjusted by Benjamini–Hochberg FDR (1).

Study 3: Variance Decomposition — H4 (Occasion dominance), E2, E3. Study 1 establishes that no dimension reaches $\text{ICC} \geq 0.50$ (H2), motivating the question of *why*. Study 3 decomposes the total variance of each HR-response metric into three sources—Person, Route, and Occasion—to test whether transient day-to-day variation is the dominant source (H4). If H4 holds, it provides a mechanistic explanation for H2: moderate ICCs arise not because the metrics are poorly defined, but because within-person session-to-session fluctuation reflects genuine physiological variability (readiness, recovery, environment) that multi-session aggregation can address. Study 3 also compares metrics to identify the most reliable candidate (E2) and examines whether rank-order consistency exceeds absolute consistency (E3).

ICC(1,1). One-way random-effects ANOVA (17) for users with $n_k \geq 5$:

$$\text{ICC}(1,1) = \frac{\text{MS}_B - \text{MS}_W}{\text{MS}_B + (n_0 - 1)\text{MS}_W}$$

$$\text{MS}_B = \frac{\sum_k n_k (\bar{x}_{k\cdot} - \bar{x}_{\cdot\cdot})^2}{K - 1},$$

$$\text{MS}_W = \frac{\sum_k \sum_j (x_{kj} - \bar{x}_{k\cdot})^2}{N - K},$$

$$n_0 = \frac{N - \sum_k n_k^2 / N}{K - 1}$$

Variance partitioning. The between-person share was $\% \text{Person} = \text{ICC} \times 100$. Within-person deviations $\tilde{x}_{kj} = x_{kj} - \bar{x}_{k\cdot}$ were predicted from the 13 route features via RidgeCV regression ($\alpha^* \in \{0.01, 0.1, 1.0, 10, 100\}$, selected by internal LOO-CV) with group k -fold CV ($k = \min(5, K_{\text{valid}})$), yielding R_{route}^2 . Because this prediction operates only on the within-person variance ($1 - \text{ICC}$):

$$\begin{aligned} \% \text{Route} &= \max(0, R_{\text{route}}^2) \cdot (1 - \text{ICC}) \cdot 100, \\ \% \text{Occasion} &= 100 - \% \text{Person} - \% \text{Route} \end{aligned}$$

The residual $\% \text{Occasion}$ captures daily physiological state, environmental conditions, sensor noise, and unmodeled route

variation. Uncertainty in the variance partitioning was estimated via user-level bootstrap ($B = 500$; 95% CI from percentiles).

Route-matched sensitivity analysis. Workout pairs were matched if:

$$\left| \frac{\Delta a_j - \Delta a_{j'}}{\Delta a_j} \right| \leq 0.20 \quad \text{and} \quad \left| \frac{A_j^+ - A_{j'}^+}{A_j^+} \right| \leq 0.20$$

where A^+ denotes total ascent ($n = 451$ pairs, 95 users). The $\pm 20\%$ tolerance accommodates GPS-derived altitude uncertainty (typically ± 10 m for barometric altimeters) and natural variation in route execution on repeat visits. Route-matched ICC provided an upper-bound estimate of person-level stability.

Split-half reliability. For users with ≥ 10 workouts, the Pearson r between the mean of the first half and the mean of the second half of chronologically ordered workouts was corrected via the Spearman–Brown prophecy formula: $SB = 2r_{sh}/(1 + r_{sh})$.

Convergence analysis. For $k = 2, 3, \dots, 10$, users with $\geq 2k$ workouts were selected. Workouts were randomly permuted within each user, then the first k and second k workouts were used to compute split-half r and SB_k . This estimates the reliability of any k randomly sampled sessions, not necessarily the first k chronological sessions of a training block; chronological aggregation could differ if fitness or route choice trends within the archive.

ICC sensitivity. ICC was recomputed with minimum-workout thresholds $n_{\min} \in \{3, 5, 8, 10, 15\}$.

Convergent validity. User-level median speed served as an external criterion; Pearson r between each metric's user-level median and median speed was computed.

Software and reproducibility. Python 3.12 (NumPy 1.26, SciPy 1.13, pandas 2.2, scikit-learn 1.5; random seed 42). Code: <https://github.com/ikari12/GRAD>.

Results

Results are organised by CRQ. Study 1 and Study 2 address CRQ1 (dimensionality); Study 3 addresses CRQ2 (variance structure) and shows how multi-session aggregation achieves reliable measurement.

CRQ1: The Speed-Based Indices Collapse to One Dimension (H1, H2). DI and FI showed $r = -0.60$ (95% CI $[-0.70, -0.48]$; bootstrap $B = 2,000$ on person-level medians), confirming H1: DI and FI capture the same underlying gradient-response coupling. Person-level medians from GPS-recorded-speed workouts alone were if anything more redundant ($r = -0.67$, 106 users) than those from Haversine-derived workouts ($r = -0.63$, 346 users); the pooled $r = -0.60$ is therefore a conservative estimate of the coupling, as Haversine speed error would be expected to attenuate rather than inflate the correlation. These subset counts are not a partition of the PCA sample and are not additive: 89 users had ≥ 5 workouts of both speed types (unique union = 363). The DI–FI correlations use all users with valid DI and FI (≥ 5

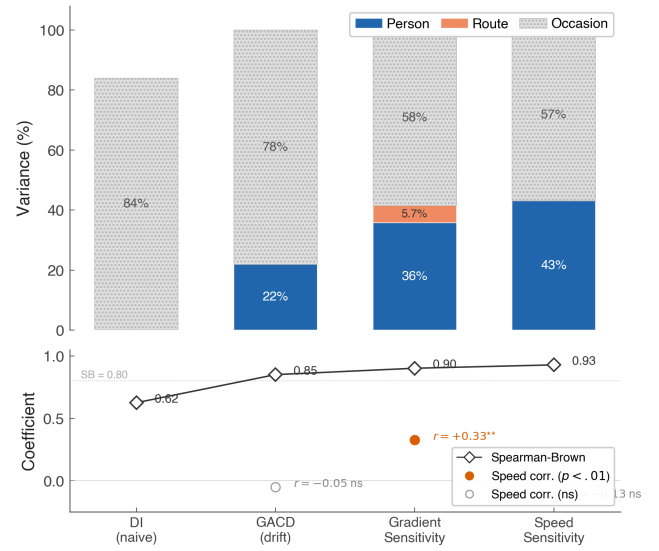


Fig. 1. The three-study durability framework. Study 1: Construct validity and effective dimensionality of speed-based indices. Study 2: Quantification and correction of the route-geometry artifact in field data. Study 3: Variance decomposition and multi-session reliability for individual profiling.

workouts; $N = 387$) and do not require RI, whereas PCA is restricted to the $N = 378$ users with all three metrics valid. The GPS-recorded subset is also smaller and more cycling-heavy than the Haversine-derived subset, so the modest difference between $r = -0.67$ and $r = -0.63$ may partly reflect subset composition rather than Haversine error alone. PCA on the $N = 378$ users with all three metrics valid retained one factor (Table 2). The second eigenvalue ($PC2 = 1.004$) fell below the parallel-analysis threshold (1.036) by 0.032 units, and bootstrap resampling placed this margin at $[-0.103, +0.011]$: a second factor was retained in only 5.3% of the 2,000 replicates, and 94.7% of replicates recovered exactly one factor. The one-factor solution is therefore not a marginal call but the stable result under resampling.

RI was largely independent of the other two indices (DI–RI: $r = +0.00$; FI–RI: $r = +0.10$; Figure 2), but its reliability was too low (ICC = 0.10; see Study 3) for this independence to be interpreted as construct distinctness rather than measurement noise. Repeating the PCA on DI and FI alone confirmed the same structure: one retained factor ($PC1 = 1.603$ vs. threshold 1.107) explaining 80.0% of variance, with DI and FI sharing 35.9% of their variance. RI is therefore treated as an exploratory, unreliable metric throughout the remainder of this paper.

To address the sport confound, stratified analysis confirmed consistent DI–FI redundancy across modalities: cycling (278 users, $r = -0.66$), running (86 users, $r = -0.59$), and mountain biking (70 users, $r = -0.72$).

The mean DI of 0.92 (Table 1) indicates that HR/Speed was lower in the second half than the first half of workouts on average—an apparent negative cardiac drift. This counterintuitive finding is explained by route geometry. Classification by ascent distribution (asc_front : fraction of total climbing in the first temporal half) reveals three route types with dramatically different DI distributions (Supplementary Figure S1):

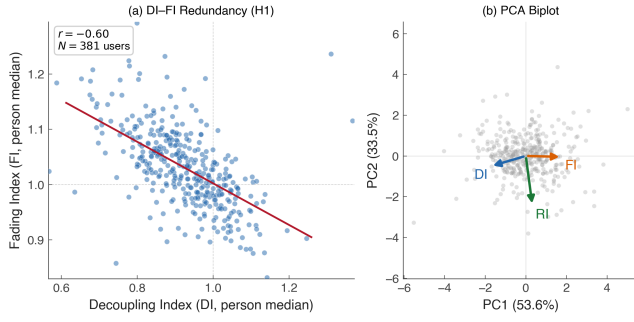


Fig. 2. Bivariate relationship between Decoupling Index (DI) and Fading Index (FI). (a) Person-level medians ($N = 378$ users with ≥ 5 workouts) show a strong negative correlation ($r = -0.60$). (b) PCA biplot: DI and FI load in opposite directions on PC1 (53.6%), while RI is orthogonal on PC2.

Table 1. Descriptive statistics of durability metrics across the included workouts ($N = 13,395$). Metric-level N is the number of workouts for which that index is defined and is therefore smaller than the inclusion total: DI requires ≥ 6 moving points per half (undefined for 200 workouts); FI requires at least one populated gradient bin in both halves (undefined for 8); RI requires ≥ 2 climbs of > 50 m (undefined for 3,715).

Metric	N	Mean	SD	Median	IQR [Q1, Q3]
DI	13,195	0.92	0.26	0.91	[0.78, 1.04]
FI	13,387	1.06	0.23	1.04	[0.95, 1.14]
RI	9,680	0.99	0.29	0.95	[0.80, 1.13]

front-climb routes ($\text{asc_front} > 0.6$; $N = 975$, 42.4%) produce higher speeds in the second half (descent), mechanically depressing DI to a median of 0.770; symmetric routes ($0.4 \leq \text{asc_front} \leq 0.6$; $N = 1,090$, 47.5%) yield a median DI of 0.978, close to unity; and back-climb routes ($\text{asc_front} < 0.4$; $N = 232$, 10.1%) show a median DI of 1.157, reflecting slower speeds in the second half. The controlled simulation (Figure 3A) directly confirms this mechanism: with cardiac drift set to zero, front-climb routes yield DI values substantially below 1.0. Additionally, warm-up effects and conservative early pacing may contribute. This has a direct practical implication: among long workouts on hilly terrain ($\Delta a > 200$ m), a large fraction of athletes are likely observing $\text{DI} < 1.0$ on their GPS platforms and incorrectly interpreting this as the absence of cardiac drift, when in fact it reflects route geometry rather than superior aerobic fitness. The 42% front-climb rate applies to this hilly inclusion set, not to GPS-tracked training in general (the filter retains 5.3% of FitRec); on flat urban sessions the geometry artifact is not the relevant problem. This observation further motivates the GACD regression, which removes the gradient confound to isolate the true physiological signal.

Table 2. PCA eigenvalues and parallel analysis on person-level medians ($N = 378$ users; Study 1). Margin is the eigenvalue minus the parallel-analysis threshold; a component is retained when the margin is positive. Eigenvalue CIs and margin CIs are percentile intervals from a user-level bootstrap ($B = 2,000$). PC2's margin CI includes zero, but a second factor was retained in only 5.3% of replicates.

	Eigenvalue [95% CI]	% Var.	Parallel 95%	Margin [95% CI]	Retain?
PC1	1.612 [1.499, 1.727]	53.6	1.152	+0.461 [+0.347, +0.576]	Yes
PC2	1.004 [0.933, 1.047]	33.4	1.036	-0.032 [-0.103, +0.011]	No
PC3	0.392 [0.297, 0.486]	13.0	0.972	-0.580 [-0.676, -0.486]	No

CRQ1: Gradient Correction Successfully Isolates Cardiac Drift (H3). The DI-FI coupling observed in H1 reflects a shared response to route gradient. Study 2 confirmed this and demonstrated that regression-based correction (GACD) successfully removes the route component, isolating true cardiac drift.

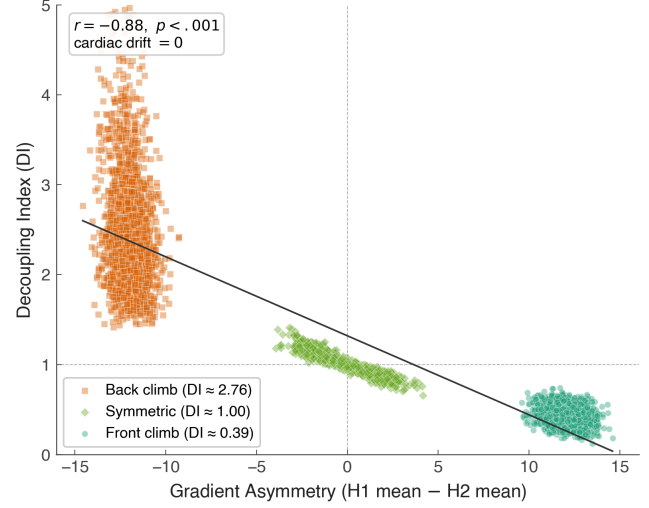


Fig. 3. Quantifying the route-geometry artifact in speed-based DI. (A) Simulation with zero cardiac drift shows that route geometry alone can produce DI between 0.70 and 1.40. (B) Empirical predictability (R^2) of naive DI vs. GACD corrected metrics.

Study 2: Correcting the Route Artifact. The GACD regression was applied to the FitRec dataset to examine whether route-geometry bias can be eliminated. In synthetic data with zero cardiac drift, route features alone predicted DI with $\text{CV } R^2 = 0.80$, the geometry-induced predictability when physiological drift is absent. Because the simulation uses a smaller gradient SD than the empirical data, this R^2 (and the 0.70–1.40 DI range) is a conservative lower bound on the artifact, not an upper bound. In real data, naive DI was predictable with $\text{CV } R^2 = 0.49$ (all routes) and 0.72 (hilly subset). After GACD correction, this predictability was eliminated ($\text{CV } R^2 = +0.02$; H3 confirmed). Unlike commercial Grade Adjusted Pace (GAP) algorithms, which typically correct pace for gradient but do not account for time-dependent cardiac drift, GACD simultaneously addresses both confounds; however, a direct head-to-head comparison with specific commercial implementations was not performed in this study. This demonstrates that GACD effectively extracts the physiological signal from the route-confounded DI (Figure 3), and explains the DI-FI coupling in H1: both metrics respond to gradient asymmetry, which GACD removes.

A natural question is whether replacing speed with estimated metabolic power—computed as $\text{speed} \times C_w(g)$ where $C_w(g)$ is the Minetti (14) walking cost polynomial—would achieve the same correction. It does not: DI/EstPower showed a route dependency of comparable magnitude but in the opposite direction ($r = -0.46$ vs. $+0.49$ for DI/Speed; $\text{CV } R^2 = 0.20$ and 0.20, respectively). On the hilly subset ($\Delta a > 400$ m), the estimated-power overcorrection was more pronounced

($r = -0.56$, $CV R^2 = 0.31$ vs. $DI/Speed r = +0.46$, $R^2 = 0.19$). The reversal occurs because the Minetti cost function drops substantially on descents ($C_w \approx 1.30$ J/kg/m at -8% and reaches a minimum of ≈ 0.94 near -15.4% , vs. 2.5 at 0%), inflating the DI denominator ratio for routes with early descents. Furthermore, the Minetti cost function was originally derived for human walking and running; its application to a dataset dominated by cycling (62.0%) introduces an inherent bioenergetic mismatch, as cycling efficiency on gradients is governed by distinct biomechanical and aerodynamic constraints. This validates the regression-based approach (GACD) as the preferred gradient correction method.

CRQ2: Multi-Session Aggregation Achieves Reliable Measurement (H4, E2, E3). CRQ1 established the supported dimensionality and demonstrated effective gradient correction. CRQ2 identifies the variance structure and shows that multi-session aggregation overcomes within-person variability to achieve reliable measurement.

Table 3. Reliability of DI and the GACD-derived metrics. β_{time} , β_{grad} , and β_{speed} are derived from Eq. 1. All four rows are computed on the same high-engagement subset ($N = 2,297$ workouts, $K = 309$ users) and by the same code path, so that DI is compared with the GACD metrics on identical data. SB is the Spearman-Brown estimate at $k = 5$ aggregated sessions, not the split-half value. Route $CV R^2$ is within-person predictability from route features (Ridge, group k -fold CV); values at or below zero indicate that route geometry carries no recoverable information about the metric once person effects are removed. Speed sensitivity may carry endogeneity bias (see Discussion); gradient sensitivity is the preferred profiling metric.

Metric	ICC(1,1)	Route-Matched ICC	SB ($k = 5$)	Route $CV R^2$
DI	0.20	0.49	0.49	-0.449
GACD (β_{time})	0.22	0.46	0.60	-0.047
Grad. Sensitivity	0.38	0.55	0.80	+0.000
Speed Sensitivity	0.45	0.75	0.89	+0.008

Occasion accounted for 56–84% of variance across all metrics (H4 confirmed; Table 3). Crucially, this within-person variability is *manageable*: multi-session aggregation progressively recovers the stable between-person signal.

Within-user route-matched ICC ($n = 451$ matched pairs, where each user's workouts were matched to their own reference route within $\pm 20\%$ of altitude range and total ascent) demonstrated this directly: holding route approximately constant increased Person estimates for all metrics—GACD ($0.22 \rightarrow 0.47$), gradient sensitivity ($0.38 \rightarrow 0.56$), speed sensitivity ($0.44 \rightarrow 0.74$)—showing that 18–31 percentage points of apparent Occasion variance is actually route-related and controllable.

Gradient sensitivity as the most reliable metric (E2). Gradient sensitivity outperformed DI and GACD across all criteria: high SB (0.80 at $k = 5$), near-zero route $CV R^2$ (+0.000), and the strongest speed correlation ($r = +0.38$). While β_{time} represents within-session drift, it should be distinguished from Maunder et al.'s (2021) "durability" construct which focuses on between-session shifts in the power–duration relationship. The ICC–SB discrepancy (DI: 0.20 vs. 0.49; gradient sensitivity: 0.38 vs. 0.80; E3 confirmed) reveals that rank-order consistency is preserved across sessions—within-person *trends* are interpretable, and multi-session averaging stabilises absolute values.

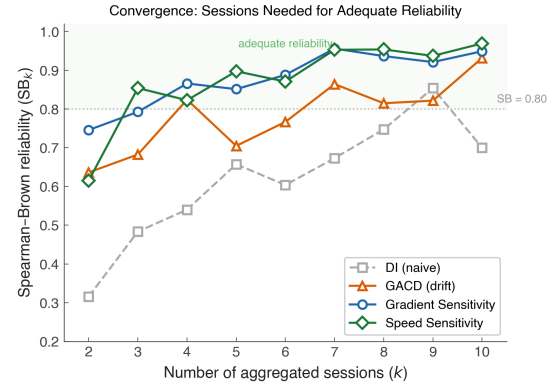


Fig. 4. Variance decomposition and multi-session reliability. (A) Variance partitioning for gradient sensitivity. (B) Convergence of Spearman-Brown reliability as a function of the number of aggregated sessions (k). Gradient sensitivity crosses the $SB = 0.80$ threshold at $k = 5$; DI and GACD do not reach $SB = 0.80$ by $k = 10$ ($SB_{10} = 0.71$ and 0.72 , respectively).

Convergence analysis (Figures 1 and 4) quantified the practical protocol: gradient sensitivity reached $SB \geq 0.80$ at $k = 5$ sessions ($SB_5 = 0.80$, $SB_{10} = 0.88$), speed sensitivity at $k = 3$ ($SB_3 = 0.81$). This provides a concrete, actionable guideline: *aggregate ≥ 5 sessions of gradient sensitivity for a reliable individual terrain-response estimate*. The median per-workout R^2 for the GACD regression was 0.44 (IQR 0.30–0.58); however, as noted in Methods, this value may be inflated by temporal autocorrelation in the GPS data and should be interpreted as an upper-bound estimate of true model fit.

Table 4. Descent placement effect on cardiac drift (GACD). Exploratory p -values are FDR-adjusted.

Analysis	Statistic	Result
Total descent (amount)	r	-0.06**
Descent-front (placement)	r	+0.16***
Within-person ($n = 108$)	Mean r	+0.082**
Steepness moderation		
Gentle ($\sigma < 5$)	r	+0.09***
Moderate (5–8)	r	+0.23***
Sport		
Cycling	r	+0.14***
Running	r	+0.18*
Mountain biking	r	+0.01 ns

Exploratory: Descent Placement (E1). A between-person association was observed between descent-front ratio and cardiac drift ($r = +0.16$, $p_{FDR} < .001$, $n = 2,297$), although it accounts for only 2.6% of between-person variance and is therefore small in practical terms. The association was moderated by terrain steepness: $r = +0.23$ on moderate terrain ($5 \leq \sigma_{grad} < 8$; $n = 735$) versus $r = +0.09$ on gentle terrain ($\sigma_{grad} < 5$; $n = 1,399$). A Fisher z test confirmed that this difference is not attributable to sampling variation ($z = 3.36$, $p < .001$). On steep terrain the association was of similar magnitude but imprecisely estimated ($r = +0.22$, $n = 163$). It was absent in mountain biking ($r = +0.01$, $n = 223$), where technical descending and variable surface

conditions plausibly dominate the HR response. Total descent amount showed a smaller association in the opposite direction ($r = -0.06$, $p = .005$), suggesting that the descent *placement* (front-loaded vs. back-loaded) matters more than the descent *amount*. Within-person evidence was modest: 65 of 106 users (61.3%) showed positive correlations, with a small mean $r = +0.089$. Seven of eight exploratory tests survived FDR correction. The between-person pattern is consistent with an eccentric-damage mechanism (16, 22), but the effect is small, the within-person support is weak, and the design is cross-sectional; this analysis is reported as hypothesis-generating rather than as evidence for the mechanism.

Discussion

From Structure to Protocol: Three Converging Lines of Evidence. The three studies form a constructive progression that builds upon, rather than contradicts, the existing theoretical framework. Study 1 showed that the speed-based indices collapse to a single empirical dimension: PCA retained one factor, and although PC2 ($= 1.004$) sat close to the parallel-analysis threshold (1.036), bootstrap resampling retained a second factor in only 5.3% of replicates. DI and FI captured a single gradient-response factor, while RI loaded on a separate dimension but with reliability too low ($ICC = 0.10$) for that separation to be distinguished from measurement error; excluding RI left the same one-factor solution. This empirical convergence of DI and FI reflects a property of the specific speed-based operationalizations used here, not a theoretical prediction of the source framework: Colosio et al. (3) argued precisely that durability and fatigability are physiologically distinct constructs that should be measured separately—a distinction that the present operationalization was designed to preserve but that the speed-based field metrics did not fully resolve. The ongoing debate in the 2025 commentaries (25–27) underscores that the question of construct distinctness remains open even at the theoretical level. Study 2 showed that regression-based gradient correction (GACD) successfully isolates true cardiac drift from route geometry ($R^2 = 0.80 \rightarrow +0.02$; H3), providing the quantitative foundation that Meixner et al.’s conceptual framework requires for field deployment. Study 3 demonstrated that multi-session aggregation overcomes the dominant day-to-day variability (56–84%; H4), with gradient sensitivity reaching adequate reliability at ≥ 5 sessions ($SB \geq 0.80$). Friel’s $< 5\%$ rule (5) represents a pioneering insight for flat or symmetric courses; GACD extends this principle to variable terrain, suggesting that Friel’s aerobic-base concept may be applicable to hilly routes where naive DI is dominated by geometry, pending prospective validation. Smyth et al. (18) demonstrated in 82,303 marathon runners that decoupling magnitude and onset predict performance, confirming the practical relevance of field-based decoupling measurement; the present study extends this line of work by identifying the route-geometry component as a major confound and providing a regression-based correction (GACD) applicable beyond the road-marathon context. The mechanism is corroborated from the opposite methodological direction by

Lloria-Varela et al. (11): under a controlled 90-minute trail protocol, uphill segments slowed progressively while HR remained stable, so that the decoupling signal tracked terrain rather than elapsed time. Their terrain-matched design removes the artifact by holding route constant, which is not an option in free-living data; GACD is the observational counterpart, removing it by covariate adjustment instead.

Construct Validity: Speed-Based DI for a Cycling-Dominated Dataset. The present analyses use HR/Speed, not HR/Power, because the majority of field users compute DI from consumer GPS watches without power meters. This choice introduces a construct validity concern for the cycling subsample (62.0% of the dataset): in cycling, speed is determined not only by gradient but also by power output, aerodynamic drag, rolling resistance, and gear selection—factors absent from running, where speed is more directly related to metabolic cost. Consequently, HR/Speed is a theoretically weaker proxy for cardiac efficiency in cycling than in running. The GACD regression partially mitigates this concern by including gradient and speed as covariates, thereby absorbing much of the terrain-related variance. However, residual aerodynamic and mechanical confounds are not modelled. Future work should either restrict primary DI analyses to the running subsample ($N \approx 1,146$ workouts) or develop a cycling-specific metabolic cost model that accounts for aerodynamic drag and drivetrain efficiency.

Whether the two-dimensional structure extends to measured power-based DI remains an open question. However, the Minetti (14) estimated-power experiment provides an informative boundary: DI computed with estimated metabolic power produced a route dependency of comparable magnitude but in the opposite direction ($r = -0.46$ vs. $+0.49$ for DI/Speed; CV $R^2 = 0.20$ and 0.20 , respectively), with the overcorrection becoming more pronounced on hilly routes ($r = -0.56$, CV $R^2 = 0.31$). The reversal occurs because the Minetti cost function drops substantially on descents ($C_w \approx 1.30$ J/kg/m at -8% and reaches a minimum of ≈ 0.94 near -15.4% , vs. 2.5 at 0%), inflating the denominator ratio. Furthermore, the Minetti cost function was originally derived for human walking and running; its application to a dataset dominated by cycling introduces an inherent bioenergetic mismatch, as cycling efficiency on gradients is governed by distinct biomechanical and aerodynamic constraints (gear ratios, reduced air resistance at lower speeds on climbs). This modality mismatch further amplifies the overcorrection, underscoring that a rigid analytical cost function is difficult to adapt to mixed-sport, real-world data—whereas the data-driven regression approach (GACD) captures modality-specific submaximal relationships directly from the signal. The key contribution of GACD is therefore not contingent on the speed-vs-power distinction: it provides a general framework for isolating cardiac drift from any terrain-confounded ratio.

A Practical Protocol for Field-Based HR Monitoring. The variance decomposition (H4) provides quantitative evidence for the IOC consensus (2) principle that external load

≠ internal response. That Occasion accounts for 56–84% of variance is not merely a consequence of uncontrolled field conditions; it is precisely the point. In controlled laboratory settings, day-to-day variation is minimised by design, creating the misleading impression that a single session suffices for reliable measurement. Real-world workouts reveal a magnitude of session-to-session fluctuation that controlled protocols cannot expose, and this provides the strongest empirical argument for the multi-session aggregation protocol proposed here. The estimate applies to the analysed sample rather than to GPS-tracked training in general: the inclusion criteria retain 13,395 of 253,020 workouts (5.3%), deliberately selecting long (> 90 min) sessions on terrain with > 200 m of altitude range. This is the regime in which route geometry can confound DI, and it is where the correction is needed; but it also means the reported Occasion share, and the prevalence of the artifact itself, should not be extrapolated to short or flat sessions. The route-matched residual (26–54%) was not explained by effort intensity ($r = -0.004$, ns), likely reflecting unmeasured readiness factors such as sleep quality, hydration, temperature, and accumulated training stress (7)—each of which represents a candidate for future integration into personalised monitoring systems.

A dual-profiling framework. The three GACD regression coefficients serve conceptually distinct roles. β_{time} (GACD) captures the within-session rate of HR drift after controlling for gradient and speed—related to, but distinct from, the between-session durability construct defined by Maunder et al. (12). In contrast, β_{gradient} and β_{speed} are not measures of time-dependent degradation; they represent route-corrected, submaximal terrain-response profiles that isolate an individual’s baseline gradient sensitivity from geometric artifacts. Based on the convergent evidence from Studies 1–3, the following dual framework is proposed: β_{time} for within-session drift description, and β_{gradient} for individual terrain-response profiling on hilly terrain. Speed sensitivity is reported for completeness but is not recommended for profiling because the speed regressor is potentially endogenous. Table 5 summarises the head-to-head comparison of the profiling metrics against DI.

Table 5. Comparison of DI and the proposed GACD-derived metrics for individual profiling, on the same high-engagement subset and code path as Table 3 ($N = 2,297$ workouts, $K = 309$ users). **DI:** Decoupling Index (HR/v); **Gradient sensitivity:** GACD coefficient for gradient; **Speed sensitivity:** GACD coefficient for speed. Route (CV R^2) values represent within-person predictability (Ridge regression on route deviations, group k -fold CV); values near zero indicate success in removing geometric artifacts. **ICC:** Intraclass Correlation Coefficient (1,1); **SB:** Spearman–Brown reliability at $k = 5$ sessions. Speed sensitivity may carry endogeneity bias (see Discussion); gradient sensitivity is the preferred profiling metric.

Metric	DI	Gradient sensitivity	Speed sensitivity
Unadjusted ICC	0.20	0.38	0.45
Route-matched ICC	0.49	0.55	0.75
SB Reliability ($k = 5$)	0.49	0.80	0.89
Route (CV R^2)	−0.449	+0.000	+0.008
Speed correlation (r)	−0.07	+0.38	−0.16

The recommended protocol is: (1) apply GACD regression to remove route-geometry artifacts from each session, (2) extract gradient sensitivity as the individual’s terrain-response

profile, and (3) aggregate ≥ 5 sessions for stable estimates. This $k \geq 5$ guideline is validated primarily in cycling (83.9% of the high-engagement subset). Sport-specific baselines are required, as gradient sensitivity varies substantially across modalities (cycling +2.11 bpm/%, MTB +1.17, running +0.47; $F = 117.6$, $\eta^2 = 0.092$). It is important to distinguish the within-session time coefficient (β_{time}) from the durability construct described by Maunder et al. (2021) (12), which defines durability as the resilience of the power–duration relationship to fatigue (a between-session or post-fatigue shift). While GACD captures the rate of internal-load drift during a session, the two constructs are related but distinct operationalizations of endurance durability.

The ICC–SB discrepancy (E3) offers additional practical insight: rank-order consistency is preserved despite session-to-session variation. This means within-person *trends* across training blocks are reproducible, even when individual sessions vary. Coaches and athletes can therefore track gradient sensitivity trends without those trends being swamped by measurement noise, provided they aggregate ≥ 5 sessions per assessment window. What reproducibility does not establish is what such a trend means: whether a change in gradient sensitivity corresponds to a change in fitness is a question of criterion validity that the present data cannot answer.

Descent Placement: Suggestive but Unconfirmed (E1). Descent placement showed a between-person association with cardiac drift ($r = +0.16$, explaining 2.6% of variance), moderated by steepness and absent in mountain biking—a pattern consistent with eccentric muscle damage (16). However, within-person evidence was small ($r = 0.089$, < 1% of within-person variance explained), and the effect sizes are too small to support firm mechanistic conclusions. The finding is therefore characterised as exploratory and suggestive.

Gradient Sensitivity: A Candidate with Known Limitations (E2, C4). Gradient sensitivity showed the most balanced profile: SB reaching 0.80 at $k = 5$, route-matched ICC = 0.55, near-zero CV R^2 , and speed correlation $r = +0.38$. The unadjusted ICC (0.38) reflects session-to-session variability addressable through aggregation; the route-matched ICC (0.55) demonstrates that controlling route selection further improves reliability.

These are reliability criteria, and it is important to be explicit about what they do and do not license. Every comparison reported here—gradient sensitivity against DI, before against after GACD correction—is an internal one: it establishes that gradient sensitivity is more reproducible than DI and less predictable from route features, not that it is a better measure of endurance fitness. The FitRec dataset contains no criterion measure of any kind (no VO_2max , lactate threshold, power output, or race result), so the claim that the residual β_{time} after correction reflects *cardiac drift* rather than some other unmeasured determinant of the HR–speed relationship—wind, drafting, surface, temperature, fuelling, pacing intent—is an interpretation, not a finding. Only the reliability and artifact-correction results are established by these data; the durability

interpretation is a hypothesis those results make worth testing.

The test we would propose is a prospective one. A cohort of endurance athletes would complete a laboratory characterisation (VO_2max , lactate thresholds, and a durability protocol such as a post-fatigue power–duration or cost-of-locomotion assessment following prolonged submaximal exercise) alongside 12 or more weeks of ordinary GPS-tracked training with power meters, sampling both flat and hilly routes. Three questions follow directly. First, concurrent validity: does aggregated gradient sensitivity ($k \geq 5$) correlate with the laboratory durability measure, and more strongly than aggregated DI does? Second, incremental validity over the correction already available to practitioners: does GACD outperform grade-adjusted pace, which normalises speed for gradient and is already deployed at scale on consumer platforms? Third, sensitivity to change: does gradient sensitivity move with a training block that measurably shifts the laboratory durability measure? A design with power meters would also allow the speed-based indices to be benchmarked directly against the HR/Power operationalization that the source framework specifies, separating the limits of the speed proxy from the limits of the correction.

Scope of Temporal Autocorrelation. The sensitivity analysis (Supplementary Figures S2a, S2b) revealed strong temporal autocorrelation in GACD residuals (median DW = 0.376; lag-1 ACF = 0.797), reducing the effective within-workout sample size by approximately 89%. It is important to delineate which analyses are affected by this finding and which are not. The autocorrelation operates at the *point level* within individual workouts; the paper’s main conclusions, however, rest on *workout-level* and *person-level* aggregated analyses. Specifically: (i) per-workout R^2 values (median 0.44) are inflated and should be treated as upper bounds of model fit; (ii) OLS standard errors for $\hat{\beta}_{\text{gradient}}$, $\hat{\beta}_{\text{time}}$, and $\hat{\beta}_{\text{speed}}$ are underestimated by a factor of $\sim 2\times$, so that under HAC correction the share of workouts with an individually significant coefficient falls to 85.0%, 59.7%, and 51.3% respectively. Per-workout inference on drift rate and speed response is therefore unreliable, while gradient sensitivity remains individually estimable in most sessions—a pattern that parallels the reliability ordering found in Study 3 and reinforces gradient sensitivity as the preferred profiling coefficient.

The claim that $\hat{\beta}$ remains unbiased deserves a qualification, because OLS is unbiased under autocorrelation only when the regressors are strictly exogenous, and the three regressors differ in this respect. Gradient is fixed by the terrain the athlete is traversing and cannot respond to a heart-rate residual, so $\hat{\beta}_{\text{gradient}}$ is unbiased on this argument; elapsed time is deterministic and likewise exogenous. Speed is not: an athlete experiencing strain may slow, which makes $v_{j,i}$ potentially correlated with earlier HR shocks and admits feedback bias in $\hat{\beta}_{\text{speed}}$, and—because speed and elapsed time covary within a session—some contamination of $\hat{\beta}_{\text{time}}$ as well. This is a further reason to prefer gradient sensitivity over the speed

and time coefficients for profiling, and it means the drift-rate coefficient should be read as a descriptive within-session slope rather than as a structural parameter. The analyses that constitute the core contributions—the H3 route-predictability test ($R^2 = 0.02$ via cross-workout cross-validation), the PCA (on person-level medians), the ICC-based variance decomposition, and the Spearman–Brown convergence curves—all operate on workout-level summary statistics (one scalar per workout) rather than on point-level residuals, and are therefore not mechanistically affected by within-workout autocorrelation. One indirect consequence is possible: if the true estimation uncertainty of $\hat{\beta}_{\text{time}}$ is larger than OLS suggests, a portion of what the variance decomposition attributes to Occasion variance (56–84%) may reflect estimation noise rather than genuine day-to-day physiological variation. This is consistent with the residual–noise conflation noted in Limitation (3), and further motivates the multi-session aggregation protocol as a way to average out both physiological and estimation noise.

Limitations. (1) *Speed-based DI*: the DI used here is HR/Speed, not HR/Power. The route artifact (C1) is specific to speed-based DI; H1, H2, H4 may differ for power-based DI. HR/Speed is a theoretically weaker proxy for cardiac efficiency in cycling than in running (see Construct Validity subsection above). (2) Secondary data with uncontrolled sensor quality; optical wrist-based HR sensors are known to have accuracy issues during exercise (7, 29, 30), and sensor noise may contribute to the high Occasion variance. (3) Residual–noise conflation: route-matched ICC partially addresses this, but full separation requires research-grade sensors. (4) No gold standard (VO_2max , power). (5) Demographics: 93.9% male, limiting generalisability. Sex differences in cardiovascular drift are well-documented, with women showing greater drift under some heat conditions; the present findings and recommendations may not generalise to female athletes. Age and training history are also unknown. (6) Exploratory findings not pre-registered. (7) Multiple comparisons: 7/8 tests survived BH-FDR, but does not account for broader analytic flexibility. (8) Observational design. (9) Single dataset. (10) Sport bias: cycling comprises 62.0% of the full dataset and 83.9% of the high-engagement subset (Studies 2–3). Cycling-specific variance decomposition (Supplementary Table S1) confirms that the Occasion-dominant pattern holds for cycling alone (Occasion 50–78% across the three GACD-derived metrics), with gradient sensitivity reaching SB = 0.93 at $k = 5$; the cycling-specific GACD SB (0.72) is lower than the pooled estimate (0.84), likely because filtering to cycling-only sessions reduces the number of qualifying users and removes between-sport variance that inflates pooled ICC. However, the running subsample in the high-engagement subset ($N = 47$ workouts from 19 users, only 3 with ≥ 5 sessions) is insufficient for stable ICC estimation; the ≥ 5 session recommendation should therefore be considered primarily cycling-specific pending validation in larger running-specific datasets. (11) Temporal autocorrelation in GACD regressions is substantial (median DW = 0.376, lag-1 ACF = 0.797; Supplementary Figures S2a, S2b): Newey–

West HAC standard errors are $\sim 2\times$ larger than OLS standard errors, and per-workout R^2 values should be interpreted as upper bounds. Under HAC correction only 85.0% (β_{gradient}), 59.7% (β_{time}), and 51.3% (β_{speed}) of individual sessions retain a significant coefficient; because the $\sim 2\times$ inflation is a lower bound, these retention rates are upper bounds. Gradient and elapsed time are exogenous to the HR residual process, but speed is not, so $\hat{\beta}_{\text{speed}}$ —and indirectly $\hat{\beta}_{\text{time}}$ —may carry feedback bias. Cochrane–Orcutt GLS on a 300-workout subsample ranked athletes similarly to OLS ($r = 0.83$ for β_{gradient}) but attenuated magnitudes, as expected when a persistent terrain regressor is quasi-differenced; OLS is therefore retained for profiling. (12) PC2 lies close to its retention threshold (margin -0.032 , 95% CI $[-0.103, +0.011]$); although only 5.3% of bootstrap replicates retained a second factor, the one-factor conclusion applies to these three speed-based indices in this sample and should not be read as evidence that the underlying theoretical constructs are unitary. (13) RI (ICC = 0.10) is an unreliable metric; its apparent independence from DI and FI may reflect measurement noise rather than true construct distinctness.

Conclusion

Under speed-based operationalization, the three constructs that could be measured (DI, FI, RI; repeatability was not operationalized) do not empirically separate (CRQ1). PCA on standardized person-level medians retained one factor: the first eigenvalue (1.612) explained 53.6% of variance, while the second (1.004) fell below the 95th-percentile parallel analysis threshold (1.036) and was retained in only 5.3% of bootstrap replicates. Restricting the analysis to DI and FI reproduced the one-factor solution (80.0% of variance). RI showed apparent independence but very low reliability (ICC = 0.10), precluding its interpretation as a distinct construct. This is a statement about the speed-based field operationalizations, not about the theoretical constructs themselves, which may well be separable under laboratory-grade measurement. Gradient-Adjusted Cardiac Drift (GACD) successfully isolates the physiological signal from route-geometry confounding, with multicollinearity remaining minimal (median VIF: gradient 2.01, speed 2.09, time 1.07). This correction reduces route predictability from 0.49 (empirical bias) and 0.80 (theoretical simulation) to near-zero ($R^2 = 0.02$; H3 confirmed). Gradient sensitivity is proposed as the preferred route-corrected profiling metric. The key to reliable field-based measurement is multi-session aggregation (CRQ2): gradient sensitivity reaches adequate reliability (SB ≥ 0.80) at ≥ 5 sessions, providing a concrete protocol for practitioners. The stable Person signal (22–44%) is recoverable through aggregation; the transient Occasion component (residual variance) remains substantial (56–84% in single sessions) but reflects day-to-day physiological variation that likely carries information about readiness and recovery. Route-matched ICC analysis showed that 18–31 percentage points of this apparent Occasion variance is actually route-related, further validating the necessity of GACD. Important caveats apply: the dataset is cycling-dominated (62.0%), and

HR/Speed is a theoretically weaker proxy for cardiac efficiency in cycling than in running; per-workout R^2 values may be inflated by temporal autocorrelation; and the dimensionality result characterises these three speed-based indices rather than the underlying constructs. Notwithstanding these limitations, integrating multi-session gradient sensitivity with daily readiness markers represents a promising direction for robust field-based fitness monitoring.

Supplementary Materials

Supplementary material is available at Figshare: <https://doi.org/10.6084/m9.figshare.32927660>. Contents: Supplementary Figure S1 (DI distribution by route type), Supplementary Figure S2a (Durbin–Watson statistics), Supplementary Figure S2b (autocorrelation function of GACD residuals), and Supplementary Table S1 (sport-specific variance decomposition).

Practical Applications

- Do not interpret speed-based DI on hilly terrain ($\Delta a > 200\text{m}$) as a cardiac-drift test. In this hilly sample, 42% of routes were front-climb profiles that mechanically produce $\text{DI} < 1.0$ with zero physiological drift.
- Prefer GACD gradient sensitivity (β_{gradient}) over DI or speed sensitivity for profiling. Speed sensitivity can be more reproducible (SB = 0.89 at $k = 5$) but may carry endogeneity bias; gradient sensitivity is the exogenous coefficient (SB = 0.80 at $k = 5$; route-matched ICC = 0.55).
- Aggregate ≥ 5 sessions. Single-session ICC is low (DI 0.20; gradient sensitivity 0.38). The $k \geq 5$ guideline is validated primarily in cycling.
- Use sport-specific baselines (cycling +2.11 bpm/%, MTB +1.17, running +0.47). Mixed-sport archives should not share a single threshold.
- Treat these as reliability and artifact-correction results, not as validated fitness diagnostics, until a prospective criterion study (VO₂max, lactate, or power-based durability) is available.

Ethics Statement

This study is a secondary analysis of the publicly available FitRec dataset (15), which contains de-identified workout telemetry (heart rate, GPS coordinates, speed, altitude) voluntarily uploaded by Endomondo users. No personally identifiable information (names, dates of birth, or contact details) is present in the dataset. The original data collection was governed by Endomondo’s terms of service and privacy policy at the time of upload. As a retrospective analysis of fully anonymised, publicly released data, this study did not require institutional ethics review under standard guidelines for secondary data research. No intervention was performed, and no individual participant can be identified from the analyses

or results presented. The dataset contains GPS coordinates that could in principle reveal frequently used workout locations; no additional spatial anonymisation was applied beyond the de-identification performed by the original dataset authors (15).

Competing Interests

The author declares no competing interests.

Funding

No external funding was received for this study.

Author Contributions

H.I. conceived the study, developed the methodology, performed all analyses, and wrote the manuscript.

Acknowledgements

The author thanks the developers and contributors of the FitRec dataset and the Endomondo community for making the telemetry data available for research.

Abbreviations

DI: Decoupling Index; **FI:** Fading Index; **RI:** Resilience Index; **GACD:** Gradient-Adjusted Cardiac Drift; **SB:** Spearman–Brown; **ICC:** Intraclass Correlation Coefficient; **VIF:** Variance Inflation Factor; **GAP:** Grade Adjusted Pace.

Data Availability

All analysis code and reproducibility scripts are available at <https://github.com/ikari12/GRAD>. The FitRec dataset is described in Ni et al. (2019) (15). Regenerating the filtered sample from the public JSON with the current scripts yields $N = 13,395$ workouts ($K = 674$ users); an earlier pipeline snapshot had reported 13,750/675. Processed feature tables (not raw GPS traces) are included in the repository.

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